Department of Economics

MSc Extended Essay Cover Sheet

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COURSE CODE	**E**	**C**	**4**	**2**	**3**

TITLE	**Gender Differences of General Election Loss on Candidate Subsequent Involvement in the UK**

WORD COUNT	**5**	**0**	**7**	**4**

Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text). Examiners are recommended to penalise excessively long essays (+ 10% of word count) by adjusting the marks by 6000/x, where x is the word count. Pages are to be numbered. Binding is not necessary, however it is crucial that all pages are secure, i.e. stapled.*

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Gender Differences of General Election Loss on Candidate Subsequent Involvement in the UK

Abstract

This essay explores cross-gender persistence among candidates contesting political positions. Using a regression-discontinuity design and the UK general election data from 2015 to 2019, I discover that losers of an election are generally likely to withdraw from subsequent races, with female affected to a lesser extent. The data does not reveal any significant deterrence gap in how candidates react to electoral loss when comparing women to men. This pattern holds true across both experienced candidates and sitting MPs. Interestingly, an initial electoral loss seems to inspire female candidates affiliated with the Conservative Party to pursue candidacy again at the 10% level.

1 Introduction

Gender inequality remains a long-term challenge in the labor market, despite improvements such as support for fertility around the world (Schönberg and Ludsteck, 2014; Landais et al., 2020). Women still face barriers in accessing leadership positions in both public and private sectors, among which political competition also serves as a typical situation, particularly in the electoral scenario. Originating from theories where Hotelling's (1929) spatial competition model, followed by Downs' (1957) expansion, suggested politicians tend to choose median voter's policy for larger chance of winning, Alesina's (1988) model proposed that female leaders may implement different policies

depending on distinct preferences. Considering that public figures must cater to societal demands, there may be variations in the political paths of men and women depending on who the voters prefer.

The importance of such differences lies in the ample evidence supporting the crucial role of women in local decision-making positions for local welfare. They bring a unique perspective to governance, emphasizing inclusivity and promoting family-friendly policies. These advantages are reflected to varying degrees in countries with different economic backgrounds, particularly evident in developing countries (Miller 2008; Brollo and Troiano 2016). A landmark study by Chattopadhyay and Duflo (2004) found that women policy makers in India were more responsive and cared more about clean water than men. Nonetheless, it is apparent that disparities persist under contemporary context in the political inclusion of women globally, though great progress has been made over the last few decades.

This raises a critical question about the underlying factors contributing to the enduring gender gap in politics. The existing literature has thoroughly investigated multiple explanations: On one hand, there exists less favourable biases against working women from party elites (Gagliarducci and Paserman, 2012). On the other hand, a shortage of women eager to stand for election and lower levels of political ambition (Schneider et al., 2016) suggests that a larger proportion of women may withdraw from competition than men, as evidenced by the lower rates of candidacy emergence observed among women.

Nonetheless, only few studies explore whether women who display initial political

ambition, as demonstrated by their candidacy for authorities, differ their propensity to persist in the subsequent entry once they lose. Consequently, mixed results appeared to be produced on whether women respond differently to electoral losses compared to men. Some researches, such as Thomsen's (2018) work on House candidates in the United States and Bernhard and de Benedicts-Kessner's (2021) study of US mayoral and state legislative elections, report no gender disparity in candidate persistence. Conversely, other studies show that female candidates are less prone to continuing their political careers after suffering defeat in elections. Wasserman (2023) observed this differential reaction primarily among novice politicians in local California elections, whereas Peveri and Sangnier (2023) indicates similar gender differences among older or former candidates in French municipal elections.

In this essay I build on this emerging literature by comparing cross-gender persistence among candidates running for political positions in the United Kingdom. Using regression-discontinuity design and voting record data from general elections (2015 –2019), I do not find any evidence supporting the existence of a gender gap in how candidates react to electoral defeat. Losers of an election tend to quit in the next election, yet such effects don’t differ across gender on their willingness to try again. Same results happen in the experienced and incumbent samples, while an initial electoral loss has an encouraging effect for female candidates who belongs to conservative party to run again significantly at 10% level. The findings remain robust when testing several bandwidth sizes around the electoral cutoff and alternative regression polynomial functions of the running variable.

By focusing on candidates running for Members of Parliament (MPs), my research will contribute to national elections competing for high-status positions in the UK context. As is analysed in House of Commons Library General Election research papers, since first being permitted to stand for election in 1918, the percentage of all women candidates has witnessed a gradual increase with the overall average of 13.8%. When examining all elections over the course of a century, Labour has consistently maintained a larger share of female candidates running in elections in comparison to rival parties. As for the number of women to sit in the House of Commons, it hit a record-breaking peak in the 2019 General Election with 220 seats, pushing the current representation level to an unprecedented 35%, far from the pre-1980s level of just 5%. Female MPs in Parliament have overwhelmingly belonged to either the Labour or Conservative Party, with these two parties being dominant for women in government.

Furthermore, the difference between the percentage of women elected and the percentage of female candidates serves as an indicator of the extent to which the increase in women candidates has translated into increased representation of women MPs in the House of Commons. During most of this period, the conversion gap has remained negative, with figures smaller than 5%. Nevertheless, from 1997 onwards, the historical trend reversed, reaching its peak in 2015 with a positive 3.4%. The 2019 General Election saw no disparity, as the proportion of women among all candidates matched precisely with the proportion of female MPs elected both standing at 33.8% (Rallings et al., 2017). The surge in female candidates in seats considered winnable post-2010 election provides a valuable opportunity to understand whether the effect of

experiencing electoral defeat varies between men and women in their participation regarding future candidacies.

This essay has important implications for the strategies adopted by political parties and women's organizations aiming to increase female representation in elected offices. Encouraging women to pursue elected positions becomes even more vital, considering that experiencing electoral defeat does not significantly deter them from running again, similar to their male counterparts. This suggests that women who aspire to political leadership roles possess a resilience that transcends gender stereotypes. By nurturing this resilience and providing support to women candidates, political parties and women's organizations can foster a more inclusive and representative political landscape. Moreover, recognizing that prominent female politicians can transcend gender in their electoral pursuits emphasizes the importance of promoting diverse leadership and amplifying the voices of women in politics. These insights can eventually contribute to building a more equitable and diverse political field where women have equal opportunities to thrive and lead.

2 Background

Although the UK has some way to go to achieve gender parity in parliament, efforts to promote gender balance in public roles and empowering female politicians never stop. According to the Inter-Parliamentary Union, the share of women in cabinet minister positions in the UK was 30% in 2023, compared with global average 23%. With women making up 35% of the House of Commons, the United Kingdom ranks

48th globally for the percentage of women holding seats in the lower house of parliament. As the fourth country to elect women to Parliament, the UK performs fairly well in comparison to other nations.

The evolution of women's involvement in British party politics has been a gradual process over time. The Parliament (Qualification of Women) Act of 1918 granted women the right to participate in parliamentary elections as candidates for the first time. As one of the countries which chose a female leader relatively early, the UK experienced the historic election Margaret Thatcher as leader of the Conservative Party in 1975 and her following tenure as Prime Minister from 1979 to 1990. Despite the shortage of related research, it appears that her era did not rise women’s representation in the party. Female leaders encountered continued attention on their gender, documented by numerous instances of both implicit and explicit sexism (O'Brien, 2015).

With the election of 120 female MPs in 1997, the proportion of seats held by women doubled, soaring to 18% of all MPs, a significant breakthrough from the era where women had never exceeded 10% of seats. The Labour Party's triumph in the 1997 general election under Prime Minister Tony Blair's leadership marked a clear transition to promoting feminist-focused initiatives. Key actions such as the adoption of all-women shortlists for parliamentary candidates were viewed as a major step towards breaking the glass ceiling. After assuming leadership of the Conservative Party in 2005, David Cameron recognized the need to challenge the Labour Party's dominance in gender equality initiatives and embarked on a mission to diversify the Conservative Party's approach to women's representation (Childs and Webb, 2011).

Under his leadership, the Conservative Party actively promoted women to prominent positions within the party and supported for policies aimed at addressing gender disparities in politics.

With 4,133 candidates, the 2010 general election set a new benchmark for the highest number of participators in electoral history. Under the preveiling atmosphere of contesting, 2010 was a turning point for gender diversity in the UK Parliament. This election witnessed a sharp increase in female candidates and election winners, marking a striking shift in the partisan composition of women's presence in political decision-making (Campbell and Heath, 2017). Moreover, this election set the stage for further progress in women's political representation in the years that followed. In the next few years, optimism about women's political representation in the UK was widespread. Various initiatives and campaigns were launched to encourage and support women in their political aspirations. For instance, the Women's Equality Party founded in 2015 aimed to challenge gender inequality in all areas of society especially politics. The party advocated for policies prioritizing the systemic barriers that women often face in pursuing political careers such as gender bias and lack of support networks.

After the 2016 EU membership referendum and David Cameron's resignation, Theresa May took the role of Prime Minister in this politically volatile environment. Her leadership during a time of political uncertainty driven by the complexities of negotiating Brexit, showed the increasing visibility and influence of women in UK politics. May's appointment, however, also brought to light the significant gender disparities that persisted within the Conservative Party. The increase in the proportion

of women MPs was not evenly distributed across parties. The Labour Party continued to lead in terms of female representation, with nearly half of its MPs being women, while the Conservative Party struggled to close the gender gap. In the meantime, the Conservative government faced criticism for rejecting proposals for more equal female representation within parliament (Elgot, 2017). During this period, the #MeToo movement spreading rapidly across the globe, had a profound impact on politics in the UK. This movement brought to light the systemic issues of sexism and harassment that many women in politics faced and sparked a broader conversation about the need for cultural and structural changes within political parties and parliament.

The resignation of Theresa May in 2019 and the subsequent appointment of Liz Truss as the UK's third female Prime Minister marked another milestone in the progression of women in British politics. Truss's appointment brought public attention on gender dynamics in top political roles and the increasing influence of women in key decision-making roles within the government (OpenAI, 2024). The 2019 general election also saw a record number of women elected to Parliament, further diversifying the composition of the House of Commons. This surge in female representation reflected growing confidence in efforts to increase women's participation in politics and signalled a shift to greater gender equality in elected office. Women from diverse backgrounds and political affiliations secured seats in Parliament, contributing to a more inclusive and representative political system.

So far the UK's journey towards gender equality in politics has been marked by both progress and challenges. In the ongoing context of public health and international

security crisis, women's voices are increasingly gaining prominence and shaping policy decisions on critical issues. Female politicians representing diverse political viewpoints, took the lead in advocating for measures to tackle urgent social and economic issues, emphasizing the value of diverse perspectives in shaping policy responses. Sustained efforts and comprehensive reforms will be essential to ensure that women have equal opportunities to participate and climb the political career ladder, facilitating the development of more effective policy measures.

3 Data and methodology

All general election results for the UK Parliament since 1918 are publicly available. I restricted the data to recent general elections that occurred on 7 May 2015, 8 June 2017 and 12 December 2019, since this period provided a relative stable sample size with readily available gender characteristics. I construct a candidate-level panel data by linking candidates to electoral constituencies and parties by matching candidates' first and last names between subsequent election cycles. I further dropped all samples that changed constituency or Party throughout their participation in general elections. Therefore, the dataset covers 8,863 observations with 6,702 individuals across 650 constituencies of England, Scotland, Wales and Northern Ireland in 2015, 2017 and 2019.

To assess the candidates' propensity to run again following an electoral failure and estimate whether differential effects exist between women and men, I conduct a regression discontinuity design on the two candidates who secured the top two highest

vote share. This yields 3,328 records, of which 1,441 are men-only races and only 359 have two women as marginal candidates. Remarkably, 1,528 mixed-gender races form a substantial proportion, accounting for 46% of the total elections.

Table 1 provides descriptive statistics for all candidates and all top two candidates with a subset of mixed-gender races. In the full sample, females who takes up 29.9 percent overall, perform better than male, with 20.4 percent succeed to be MPs, 37.5 percent continue to compete in general election next time, and 38.4 percent run again in the next two elections. These figures rise modestly in the limited sample of marginal candidates and its subset due to restriction on losing candidates who are in second places and further mixed gender races only.

Indeed, despite the increasing proportion of female entering top two rankings, the percentage of MPs elected from female candidates turns out to be no longer dominant relative to their male counterparts when moving to a smaller marginal candidate pool. Persistence in the future elections follows a similar pattern. The higher fraction of women remains in the re-entry choice among marginal groups, whereas the corresponding gender gap becomes tiny below 1 percent. In the subsample of mixed-gender races, this pattern reverses that men tend to exhibit a stronger inclination to try again compared to women. These summary statistics suggest that male and female candidates may have varying responses to an initial electoral defeat when deciding whether to pursue further involvement in politics.

Table1: Summary statistics

	All candidates		Marginal candidates		Marginal candidates in mixed-gender races	
	Female	Male	Female	Male	Female	Male
Female	29.92%		34.13%		50.85%	
Elected in t	20.40%	19.58%	47.62%	55.47%	45.17%	60.59%
Persist in the next GE	37.54%	33.98%	61.86%	61.06%	60.69%	62.03%
Persist in the next two GEs	38.38%	34.74%	62.15%	61.26%	61.10%	62.24%
Characteristics						
Previous candidates	31.00%	34.89%	48.50%	61.59%	48.01%	66.44%
Sitting MPs	17.42%	18.69%	40.67%	52.65%	38.87%	58.72%
Labour Party	28.09%	15.79%	56.78%	35.26%	59.59%	27.30%
Conservative Party	15.12%	18.34%	28.87%	46.17%	24.71%	55.39%
Observations	2,652	6,211	1,136	2,192	777	751
Total Observations	8,863		3,328		1,528	

Through my analysis, I explore the gender-related variations in future electoral entry among the candidates who narrowly succeeded in winning or miss out on victory in each election round. The following regression is the baseline specification separately for men and women:

$$Y_{i,j,t+2} = \alpha + \beta Fail_{i,j,t} + f\left(Margin_{i,j,t}\right) + Fail_{i,j,t} \times f\left(Margin_{i,j,t}\right) + \gamma_t + \delta_j + \eta_i + \varepsilon_{i,j,t}$$

where $Y_{i,j,t+2}$ is outcome variable of interest, which equals to 1 if candidate i in t = 2015, 2017 recontests in constituency j in the next election $t+2$ = 2017, 2019, and 0 otherwise, $Fail_{i,j,t}$ is treatment indicator that equals 1 for candidate i failed to win in the initial election t and 0 otherwise, and the variable $Margin_{i,j,t}$ functions

as the running variable, representing the margin of victory or loss, which is the distance index between the vote share of the winner and the first runner-up. The coefficient β denotes the effect of failure in the initial election on next political involvement.

The parametric RD model is estimated by fitting local linear functions $f(Margin_{i,j,t})$ in the neighbourhood of the threshold with uniform kernel weights at optimal bandwidth. For completeness, I also explore the sensitivity of varying bandwidths, a second order of polynomial, non-parametric approaches with suggested bandwidths and full unrestricted sample. I also control for election year fixed effects γ_t, constituency fixed effects δ_j and party fixed effects η_i. Considering the correlation in outcomes across individuals within the same constituency cell, the standard errors are clustered based on the constituency level.

To further assess whether candidates exhibit gendered responses to loss, I conduct the second specification adding gender and all interaction terms between genders and winning position as follows:

$$\begin{aligned} Y_{i,j,t+2} = \alpha + \beta_1 Fail_{i,j,t} + \beta_2(Fail_{i,j,t} \times Female_i) + \beta_3 Female_i + f(Margin_{i,j,t}) \\ + Fail_{i,j,t} \times f(Margin_{i,j,t}) + Female_i \times f(Margin_{i,j,t}) \\ + Fail_{i,j,t} \times Female_i \times f(Margin_{i,j,t}) + \gamma_t + \delta_j + \eta_i + \varepsilon_{i,j,t} \end{aligned}$$

where all other variables are defined as above and β_2 now represents the difference in impact of failure for female candidates compared to male.

4 Empirical analysis

Figure 1 presents a visual insight into how failure impacts future political commitment. The graph compares the likelihood of engagement in the next election $t+2$ with the margin of victory in election (year t) within bin, with a 2nd-order local polynomial fit plotted on either side of the zero threshold.

Panel A shows all marginal candidates aggregated. On the left side of the threshold, it can be observed to be a relatively smooth U-shaped correlation between candidates' margin of victory and their probability of seeking re-election. Conversely, on the right side, successful candidates exhibit a quadratic form, with individuals over the margin of victory at 0.3 experiencing a decline in reemergence. An abrupt discontinuity arises at the threshold, with an almost 50 percentage point increase. The jump highlights the discouragement resulting from defeat, statistically different from zero at 1% level.

The graphs in panel B and C present individual scatter plots and local polynomial fits for female and male candidates respectively. Compared to male candidates, the trend for female candidates is more curved: female losers initially decrease to the margin of victory at about -0.4 before rising, while female winners increase slightly and then subsequently declining. Though these jumps are statistically significant, the increase in deciding to re-enter induced by the cutoff is smaller for women (37 percentage points) compared to men (52 percentage points).

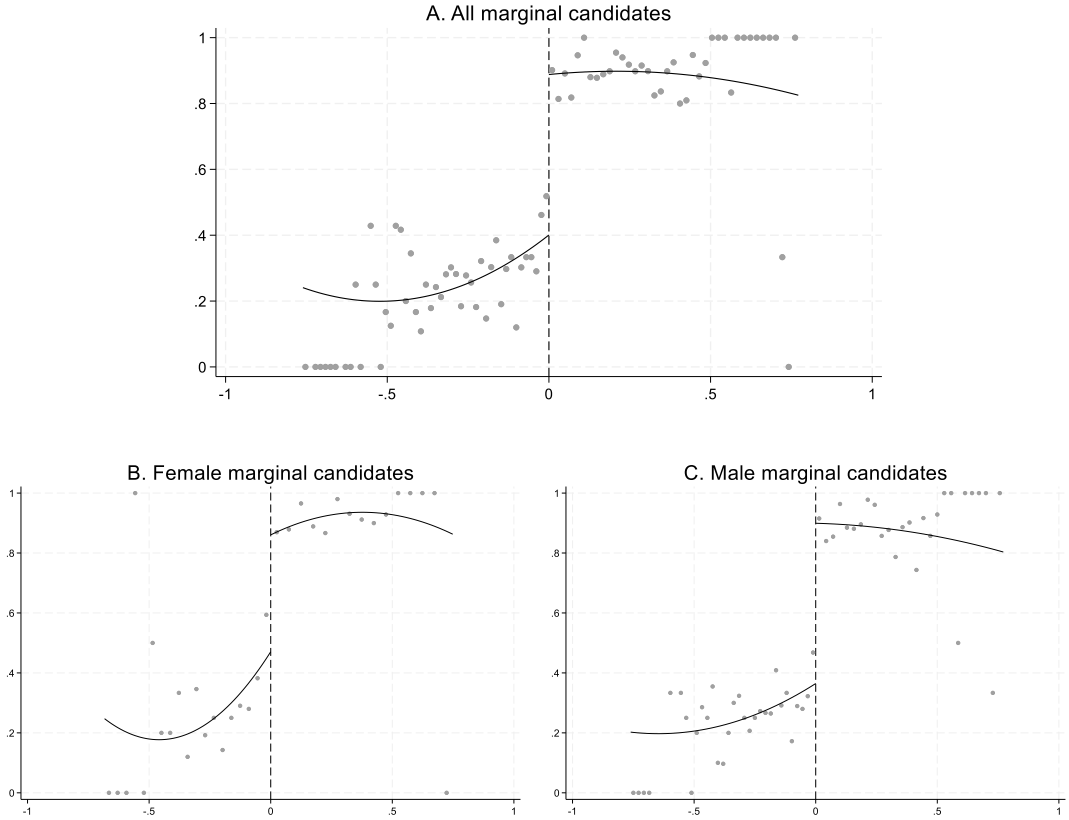

Figure 1: The discontinuity in candidate reemergence at the cutoff

Note: Each point is a local sample average within bin. Bins are selected using evenly-spaced mimicking Variance. The solid line is regressed with a polynomial fit of order 2 on each side separately with uniform kernel weights. Full sample are listed at the top with subsamples of female and male separately below.

Table 2 reports the baseline results using the first equation, with all top two candidates and female and male candidates in separation. Columns 1-5 are parametrically estimated as explained earlier, except for the non-parametric model (column 4), which allows for more flexible functional form assumptions (Calonico et al., 2014). In the first three columns, a local linear model with optimal bandwidth is centred around the cutoff, using. A second order polynomial fitted above and below the threshold is applied in column 5. Column 6, including the entire pool of marginal candidates without constraints on the margin of victory range, may have serious

endogeneity problems.

Overall, losing an election has a negative and statistically significant impact on candidates' determination, although column 2 and 5 in panel B stand out as exceptions. Across all specifications in panel A, the defeat in an initial election leads to a decrease of 35 to 52 percentage point in the willingness to persist in two years at 1% significance level. With approximately 89% of victorious candidates deciding to run again, the 45 percentage points drop denotes a 50 percent discouragement effect of losing.

Panels B and C reveal similar outcomes across both male and female candidates without significant deviation. While the propensity of both female and male winners to engage in future general elections diminishes, the impact of electoral defeat is more substantial for men than for women in terms of magnitude. At the suggested mean squared error optimal selector in the fourth column, when facing defeat in a general election, women candidates are expected to experience a 48-percentage point decline in the probability of seeking future candidacy, while men are expected to experience a 51-percentage point decline.

The distinction could be explained by the trend where women engaged in closely fought elections tend to exhibit marginally higher rates of reemergence as candidates regardless of their election outcome, in contrast to men. It is expected that female close winners will have a predicted probability of running again at 90 percent, while male close winners are anticipated to have a probability of 89 percent. In contrast, female close losers are expected to have a probability of 42 percent, and male close losers are expected to have a probability of 38 percent.

Table 2: Effect of electoral defeat on subsequent involvement

	Polynomial order one				Polynomial order two	
	(1) Optimal bandwidth	(2) 0.5*optimal bandwidth	(3) 2*optimal bandwidth	(4) Non-parametric	(5) Optimal bandwidth	(6) Full sample
Panel A. All marginal candidates						
Fail	-0.446***	-0.349***	-0.518***	-0.432***	-0.360***	-0.500***
	(0.0672)	(0.0946)	(0.0476)	(0.0766)	(0.0922)	(0.0549)
Winner mean	0.893	0.879	0.895	0.892	0.893	0.893
Bandwidth	0.198	0.099	0.397	0.164	0.198	-
Observations	943	520	1,885	801	943	2,202
R-squared	0.601	0.646	0.596	0.621	0.605	0.590
Panel B. Female marginal candidates						
Fail	-0.458**	-0.470	-0.436***	-0.479**	-0.424	-0.381**
	(0.194)	(0.290)	(0.143)	(0.231)	(0.271)	(0.161)
Winner mean	0.902	0.885	0.910	0.899	0.902	0.913
Bandwidth	0.225	0.112	0.449	0.188	0.225	-
Observations	334	181	623	286	334	679
R-squared	0.792	0.867	0.795	0.806	0.792	0.796
Panel C. Male marginal candidates						
Fail	-0.524***	-0.504***	-0.556***	-0.511***	-0.473***	-0.528***
	(0.0704)	(0.102)	(0.0625)	(0.100)	(0.101)	(0.0788)
Winner mean	0.890	0.897	0.883	0.893	0.890	0.884
Bandwidth	0.385	0.192	0.770	0.195	0.385	-
Observations	1,259	622	1,522	635	1,259	1,523
R-squared	0.654	0.647	0.641	0.651	0.654	0.641

Notes: * p<0.10, ** p<0.05, *** p<0.01. Standard errors clustered by constituency between brackets.

Table 3 presents the results from estimating the second equation, indicating that losing in the initial election has a negative effect on male candidates' next-time reentry at a significance level of at least 5%, and the role of gender has a positive but insignificant effect. Male candidates face a substantial decline of 33 to 54 percentage points in the likelihood of running in another election within two years following an initial electoral defeat. The coefficient on the interaction term $Fail_{i,j,t} \times Female_i$

reveals that female candidates, when compared to their male counterparts, tend to be -7 to 13 percentage points less likely to pursue further engagement after experiencing an initial electoral loss.

Compared to Bernhard and de Benedicts-Kessner's (2021) findings in which men who lose for US state legislative elections are 38 percentage points less inclined to seek further participation relative to women (39 percentage points), the gender gap in my results is a bit larger. Taking the estimates from the first column, there is a 13-percentage point differential in this gap between genders. This suggests that the impact of electoral failure is less discouraging for female candidates, with a 26 percent decrease in deterrence effect over the male among winners, but not significant at conventional levels.

Table 3: Gender effect of electoral defeat on subsequent involvement

	Polynomial order one				Polynomial order two	
	(1)	(2)	(3)	(4)	(5)	(6)
	Optimal bandwidth	0.5*optimal bandwidth	2*optimal bandwidth	Non-parametric	Optimal bandwidth	Full sample
Fail	-0.489***	-0.326**	-0.540***	-0.454***	-0.362***	-0.543***
	(0.0887)	(0.126)	(0.0624)	(0.0990)	(0.121)	(0.0715)
Fail*Female	0.126	-0.0726	0.0636	0.0650	0.00263	0.129
	(0.165)	(0.244)	(0.118)	(0.186)	(0.219)	(0.133)
Female	-0.00392	0.0396	0.00564	0.0435	0.0922	-0.0428
	(0.116)	(0.187)	(0.0822)	(0.134)	(0.151)	(0.0907)
Winner mean	0.893	0.879	0.895	0.892	0.893	0.893
Bandwidth	0.198	0.099	0.397	0.164	0.198	-
Observations	943	520	1,885	801	943	2,202
R-squared	0.603	0.651	0.597	0.622	0.608	0.592

Notes: * p<0.10, ** p<0.05, *** p<0.01. Standard errors clustered by constituency between brackets.

5 Heterogeneity

To explore possible correlation between electoral setback and the return of candidates, I take a closer look at subsamples by splitting along observable characteristics such as prior experience, incumbency status or party allegiance. Table 4 is the heterogeneity analysis using local polynomial specification with optimal bandwidth, indicating that the overall effect of encountering an electoral loss on the determination remains negative, and differential deterrence of women relative to men exists on different subsamples conditioned on different variables.

Starting by examining whether difference in previous experience of election across gender are reflected in marginal candidates, it is shown that there are negative effects on future political participation of former male candidates at 1% significance level, and consistently positive differential deterrence of female. For those in their first time competing, the model with the 2nd-order polynomial gives a more discouraging deterrence of female newcomers. No significant gender gap among both groups implies that the patterns of variation across gender cannot be attributed by accumulated experience.

Moving to the comparison between sitting MPs and challengers, it is clear that there are stronger negative effects of failure on the decision at the point of reentry of male incumbents than male challengers, which provides new evidence against Lee(2008) highlighting potential disadvantages for challengers in U.S. House, and Baskaran and Hessami (2022) arguing no gap based on local council elections in Germany. The deterrence effect remains insignificant yet in opposite direction in two

groups in both models. Compared to male candidates, female incumbents who were sitting in the position enjoy a weaker deterrence effect ranging from 29 to 48 percentage points after an initial loss. In comparison, female challengers who decided to compete face a more severe deterrence effect ranging from 10 to 43 percentage points.

Regarding party allegiance, I picked two largest parties in the UK, Labour and Conservative, as the representatives considering their sample size. As is shown below, the effect on the re-contesting decision for Conservative party's male candidates remains negative, in contrast to the insignificant figures for Labour sample. Labour candidates have stronger negative deterrence of women over men when she failed to win in the initial election, but such pattern got reversed in Conservative party. Particularly when taking the estimates from the model with the second-order polynomial for samples belongs to conservative party, it is surprising to see that the coefficient on the interaction term represents that an initial electoral loss now has an encouraging effect of 11 percentage points (or 71 percentage points larger) for female candidates to run again, which is statistically significant at 10% level.

Looking at the significant gap between men and women, it is clear that the Labour Party has been advancing gender equality, mitigating gender biases in candidate selection. However, the Conservative Party, as is discussed in the background before, still struggles with gender tensions and fails to shift the focus away from gender. This environment fosters increased courage and a positive competitive mindset among women who experience frustration in the face of gender discrimination.

Table 4: Heterogeneity

	Panel A. Previous experience		*Panel B. Incumbency status*		*Panel C. Party allegiance*	
	(1) Former candidates	(2) New candidates	(3) Incumbents	(4) Challengers	(5) Labour	(6) Conservative
Optimal Bandwidth	0.293	0.187	0.205	0.171	0.185	0.388
I. Polynomial order one						
Fail	-0.590***	-0.687**	-0.723***	-0.526*	-0.469	-0.596***
	(0.125)	(0.311)	(0.183)	(0.273)	(0.331)	(0.155)
Fail*Female	0.118	0.137	0.286	-0.0960	-0.142	0.419
	(0.241)	(0.494)	(0.379)	(0.460)	(0.471)	(0.270)
Female	0.126	-0.0478	0.117	0.0565	0.282	0.121
	(0.168)	(0.351)	(0.282)	(0.326)	(0.360)	(0.300)
Winner mean	0.889	0.937	0.876	0.928	0.856	0.904
Observations	808	367	483	411	358	712
R-squared	0.723	0.901	0.853	0.802	0.774	0.825
II. Polynomial order two						
Fail	-0.537***	-0.322	-0.768***	-0.462	-0.0294	-0.597***
	(0.186)	(0.457)	(0.264)	(0.436)	(0.536)	(0.223)
Fail*Female	0.0447	-0.516	0.484	-0.434	-0.475	0.711*
	(0.365)	(0.740)	(0.511)	(0.716)	(0.719)	(0.370)
Female	0.322	0.280	0.0922	0.218	0.569	0.0262
	(0.220)	(0.498)	(0.370)	(0.449)	(0.502)	(0.328)
Winner mean	0.889	0.937	0.876	0.928	0.856	0.904
Observations	808	367	483	411	358	712
R-squared	0.726	0.908	0.854	0.804	0.789	0.829

Notes: * p<0.10, ** p<0.05, *** p<0.01. Standard errors clustered by constituency between brackets.

6 Robustness check

To validate the RD design, I check the smooth distribution across the threshold and balanced covariates as the margin of victory converges towards 0. In addition, I relax the definition of dependent variable as future electoral entry within the next two elections for the robustness check.

Under the framework of this essay, it is unlikely to be the case for the candidates to manipulate electoral results in the UK context since the MPs general election adopts the first-past-the-post system, a single-member plurality approach to electoral representation. Figure 2 panel A displays the density of margin of victory among all top two candidates, with the graphs for women and men in panels B and C respectively, showing no discontinuities in the vicinity of the cutoff when the electoral race is tight. Therefore, the hypothesis that the running variable is not subject to manipulation is known to be satisfied. Testing the density around the cutoff developed by Mccrary (2008) returns a p-value of 0.816 among full sample (p-value of 0.304 for female only and 0.994 for male only), and further rejects the existence of a statistically significant leap in the running variable at the discontinuity point.

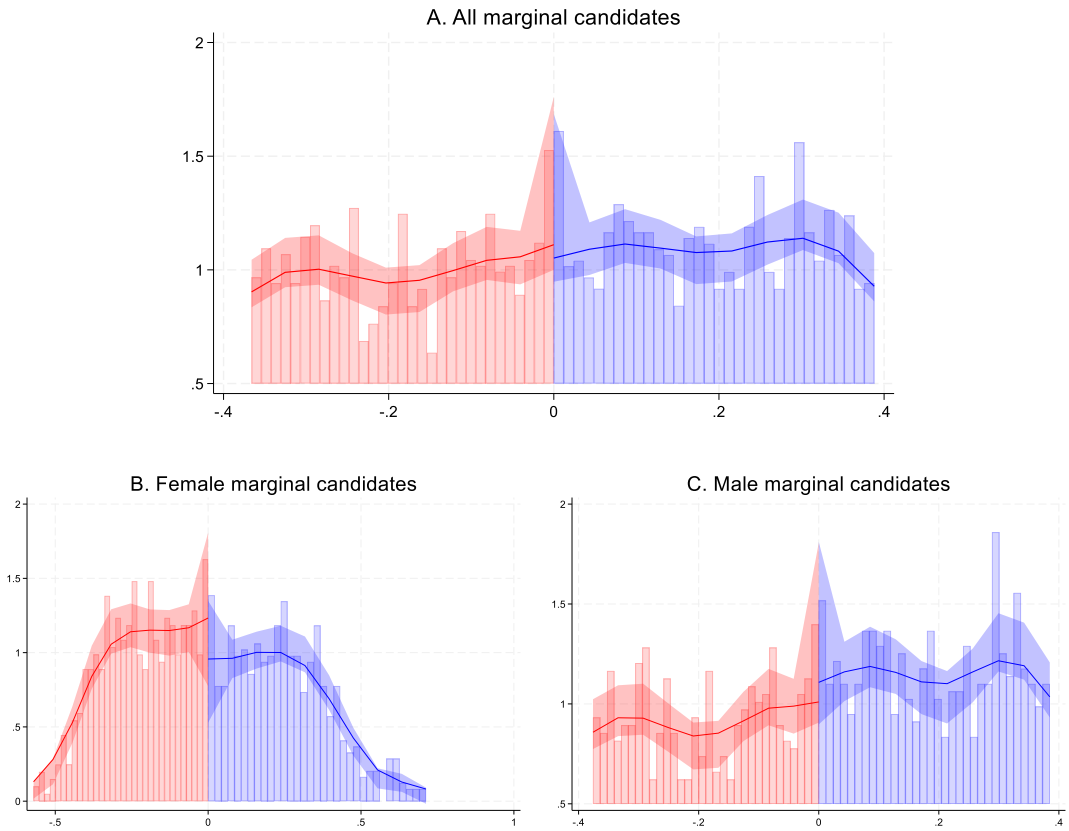

Figure 2: Distribution of margin of victory

Note: The solid line illustrates a local polynomial distribution of the running variable with triangular kernel weights. The shaded areas are confidence bands at 95% level. Full sample is listed at the top with subsamples of female and male separately below.

Next, I proceed to perform covariate balance tests to demonstrate that there are no significant changes in indicator characteristics around the victory cutoff. Binned scatterplots for total votes, previous experience and incumbency status are displayed in Figure 3 plotted on the running variable.

As is illustrated in the first row for the pooled sample, samples close to the cutoff show no potential jumps at the threshold. Regressing each covariate on margin of victory yields the p-values of 0.896, 0.588 and 0.810 respectively, indicating that the hypothesis that there is no breakpoint cannot be rejected and it can be seen that the local

continuity hypothesis is satisfied.

Panels A1 to C3 list the differences in these characteristics for the groups of female and male candidates. The graphs imply that female candidates who narrowly win or lose elections differ from male candidates facing similar outcomes. Compared to women, men are more inclined to have past electoral candidacy experience and currently hold incumbent positions. On the other hand, they are similar in terms of total votes in close elections.

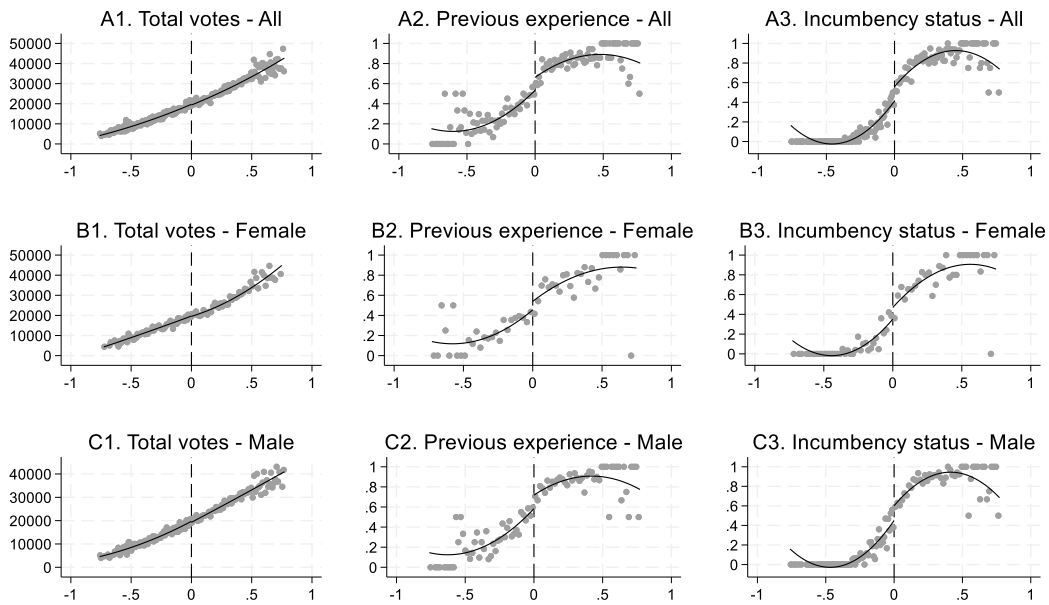

Figure 3: Balanced covariates

Note: 2nd-order polynomial is fitted on each side of the cutoff. Full samples are listed at the top with subsamples of female and male separately below.

To further check the robustness of the main findings, I test whether different interpretation of persistence influences the outcomes. In the baseline specifications, I specify the dependent dummy variable to be one when a candidate with the same full

name is listed on both the identical constituency and party only for the upcoming election in two years. If a candidate in 2015 waits until 2019 election as a second participation, then he or she would be labelled as not seeking reelection. In order to account for this potential bias, I introduce a modified dependent dummy variable, which is set one if a candidate appears on any of the future two elections and re-run the baseline regressions.

Table 5 outlined the outcome, which is consistent with baseline estimates with a gap of -8 to 14 percentage points in deterrence effect. For candidates identified as persistent over the next four years, the impact of a general election loss on candidate reemergence seems to be slightly exacerbated among women candidates. Nevertheless, there is not compelling evidence indicating substantial divergence in behaviour between female winners and losers compared to their male counterparts.

Table 5: Robustness check

	Polynomial order one				Polynomial order two	
	(1) Optimal bandwidth	(2) 0.5*optimal bandwidth	(3) 2*optimal bandwidth	(4) Non-parametric	(5) Optimal bandwidth	(6) Full sample
Fail	-0.478***	-0.336**	-0.538***	-0.462***	-0.375***	-0.535***
	(0.0916)	(0.129)	(0.0633)	(0.101)	(0.125)	(0.0713)
Fail*Female	0.135	-0.0810	0.0780	0.106	0.0584	0.133
	(0.182)	(0.262)	(0.124)	(0.198)	(0.235)	(0.138)
Female	-0.00471	0.0241	0.0118	0.0294	0.0605	-0.0339
	(0.124)	(0.198)	(0.0841)	(0.136)	(0.158)	(0.0918)
Winner mean	0.897	0.897	0.896	0.890	0.897	0.893
Bandwidth	0.191	0.095	0.382	0.171	0.191	-
Observations	909	499	1,830	818	909	2,202
R-squared	0.595	0.642	0.591	0.605	0.599	0.589

Notes: * p<0.10, ** p<0.05, *** p<0.01. Standard errors clustered by constituency between brackets.

7 Conclusion

The upward trend in the number of women stepping forward as candidates seems to be a promising signal for fostering greater women's representation around the world. As one of the countries which realized the importance of gender balance in politics and public life early, the UK achieved great progress compared to other countries. This places the UK in a unique position to provide valuable insights at the national level into the development of these candidates along their political path. I therefore draw on the candidate-level dataset from recent three general elections for the UK Parliament between 2015 and 2019 alongside a causal RD design to evaluate the varying effects of electoral loss on the inclination of male and female candidates to contest future elections.

My analysis focusing on close election matchups reveals that the consequence of suffering a loss in a general election on the persistence in politics appears to be somewhat reduced for female candidates compared to males. It seems that women who barely lose elections are marginally more likely to seek future candidacies than their male counterparts in similar circumstances. The difference in effect between male and female candidates is negligible. These findings remain robust and hold true across both experienced candidates and sitting MPs. Interestingly, however, for female candidates affiliated with the Conservative Party, an initial electoral setback appears to foster a renewed determination to run again, a phenomenon observed at a significant level of 10%.

One possible explanation for the higher reelection rates among women in political

careers could stem from a positive selection process in their decision to pursue such paths. The tremendous challenges in breaking through traditional gender stereotypes and societal expectations gives rise to a group of exceptionally competent and capable female politicians. As a result, these women are more inclined to have expertise in navigating the landscape of political governance. This phenomenon may result from unobserved political talents possessed by these women, which in turn enhance their electoral achievements and prospects for reelection.

Encouraging more women to pursue political candidacy appears to have minimal drawbacks, considering that women who experience electoral defeat are just as inclined to run again as their male counterparts. This finding challenges any assumptions of inherent gender-based deterrence in political participation. It suggests that promoting female candidates not only fosters gender equality but also contributes to a more resilient and inclusive political landscape. Additionally, the observation that prominent female politicians may transcend gender in their candidacy decisions underlines the importance of fostering a political environment where merit and capability are valued above gender stereotypes. Therefore, strategies focused on empowering and supporting female candidates regardless of electoral outcomes could play a crucial role in achieving greater gender parity in political leadership.

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